

Extending Kubernetes Self-Healing for Networking Failures

CS331 - Computer Networks Project

Arpan Gupta (24110051)

Buha Deep Maheshbhai (24110082)

Param Tanna (24110236)

Ramji Purwar (24110287)

Rathod Yuvraj Rajubhai (24110293)

Solanki Viraj Rajeshbhai (24110348)

Problem Statement & Objective

Problem: Kubernetes heals container crashes but is **blind to network-layer failures**.

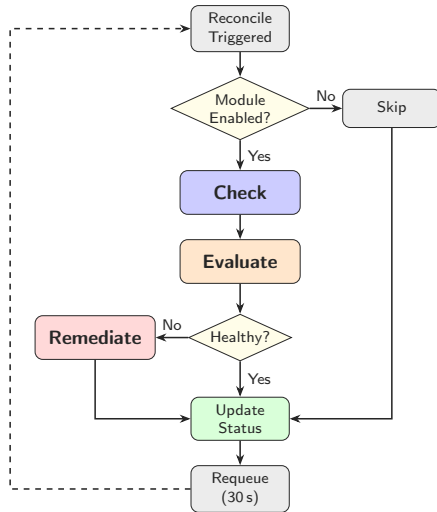
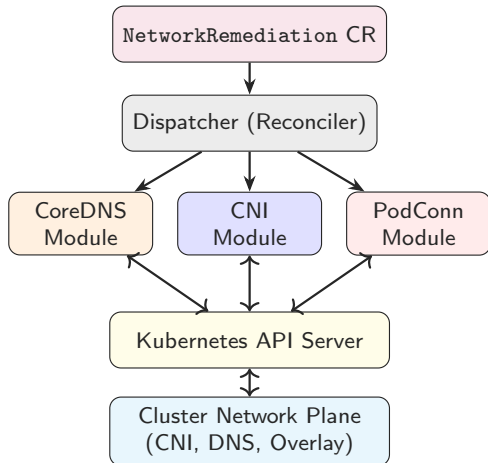
- ▶ CNI plugin crashes → pods stuck in `ContainerCreating`
- ▶ CoreDNS degradation → silent DNS resolution failures
- ▶ Overlay tunnel drops → cross-node traffic silently fails
- ▶ Pods show `Running` (1/1 `Ready`) while network is dead

Objective

Build a **custom Kubernetes operator** that continuously monitors and auto-remediates three critical network subsystems:

1. **CNI plugin health** (Calico IPAM exhaustion & agent failures)
2. **CoreDNS availability** (scale-to-zero, corrupted upstreams, latency spikes)
3. **Pod-to-pod connectivity** (overlay tunnel drops & cross-node routing)

System Architecture & Design



Extension: Custom Kubernetes Operator

Operator Architecture & Control Plane:

- ▶ **Custom Go Operator:** Built with Kubebuilder; extends Kubernetes self-healing directly into the network data plane.
- ▶ **Cluster-Scoped CRD:** Single declarative `NetworkRemediation (v1alpha1)` resource governing all monitoring and remediation.
- ▶ **Modular Subsystem Toggles:** Independent enabled flags for CNI, CoreDNS, and Pod Connectivity modules with zero coupling.
- ▶ **Three-Phase Pipeline:** Standardized *Check* → *Evaluate* → *Remediate* reconciliation executed every 30 s.

Core Capabilities & Safety Guardrails:

- ▶ **$O(N)$ Ring Probing:** Adapted Pingmesh topology ($2N$ probes vs. $O(N^2)$ full mesh) with intra-node canary validation.
- ▶ **Configurable CRD Specs:** Users can define module specifications and detection thresholds directly in the CRD.
- ▶ **Safety Guardrails:** 30 s cooldown windows, consecutive failure thresholds ($N \geq 3$), and bounded restart escalation.

Issues Fixed

CNI Module

- ▶ calico-node agent crash
- ▶ Disabled Calico IPPool
- ▶ IPAM pool exhaustion

CoreDNS Module

- ▶ CoreDNS replicas scaled to zero
- ▶ Corrupted upstream forward directives
- ▶ CPU throttling (≤ 5 m) latency spikes

Pod Connectivity Module

- ▶ Downed virtual interface (veth)
- ▶ Broken IPIP overlay tunnel (tun10)

Non-Functional Testing Parameters

Performance & Reliability:

- ▶ **Remediation time:** 30-60 s across all modules
- ▶ **Reconcile interval:** 30 s (configurable)
- ▶ **Cooldown windows:** 30 s per remediation type to prevent restart thrashing
- ▶ **Consecutive failure threshold:** 3 (avoids reacting to transient issues)

Safety & Resource Efficiency:

- ▶ **Independent toggles:** each remediation type gated separately
- ▶ **Single operator pod:** manages all three modules
- ▶ **Minimal footprint:** no additional DaemonSet or sidecar required

Challenges Faced

- ▶ **Minikube & Kubebuilder Setup:**

- ▶ Multi-node Minikube + Calico CNI setup.
- ▶ Custom RBAC and CRD scheme registration in Kubebuilder.

- ▶ **Understanding Pod-to-Pod Networking:**

- ▶ Layered data path: veth pairs, bridges, and tunl0 overlay.

- ▶ **Failure of Reconcile Loop Cooldown:**

- ▶ Duplicate restarts before network convergence.
- ▶ Resolved via 30 s cooldowns and sequential pipeline execution.

Thank You!

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